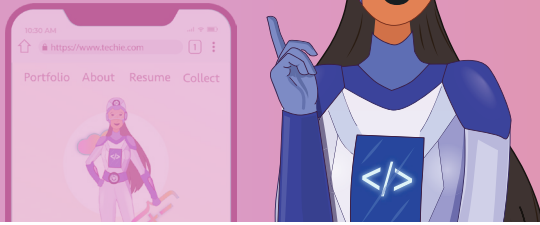


Links and Forms

09



A personal website is a great way to showcase your skills and projects with everyone.



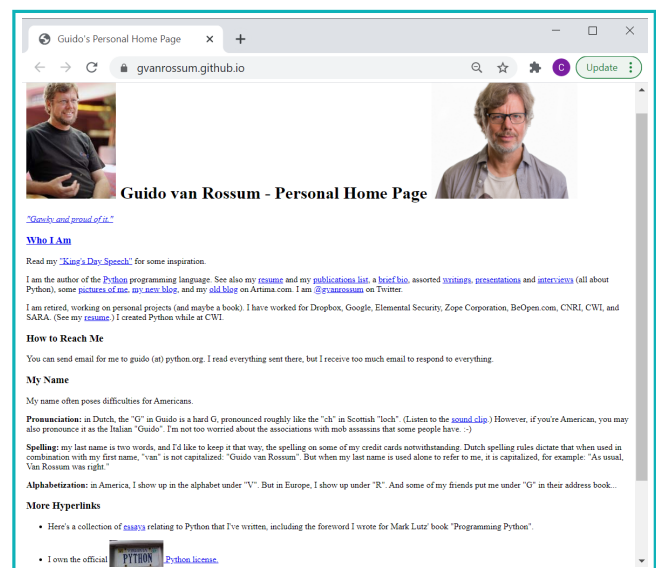
In the previous lesson, we learnt how data on the internet can be accessed through websites. Websites are a great platform to showcase information about us, our skills, achievements and projects in a structured way. This kind of a website is called a Portfolio.

Portfolio

A portfolio is a collection of works or documents which demonstrate a person's skills and accomplishments. A photographer's portfolio of images and an artist's portfolio of artwork are types of portfolios.



For example: Here is a sample portfolio of a world-famous programmer and inventor of the Python programming language, created using simple HTML.



A Portfolio site can be considered an extension to a personal page or resume. Someone with a well-designed portfolio will stand out when they enter competitions or apply for jobs.

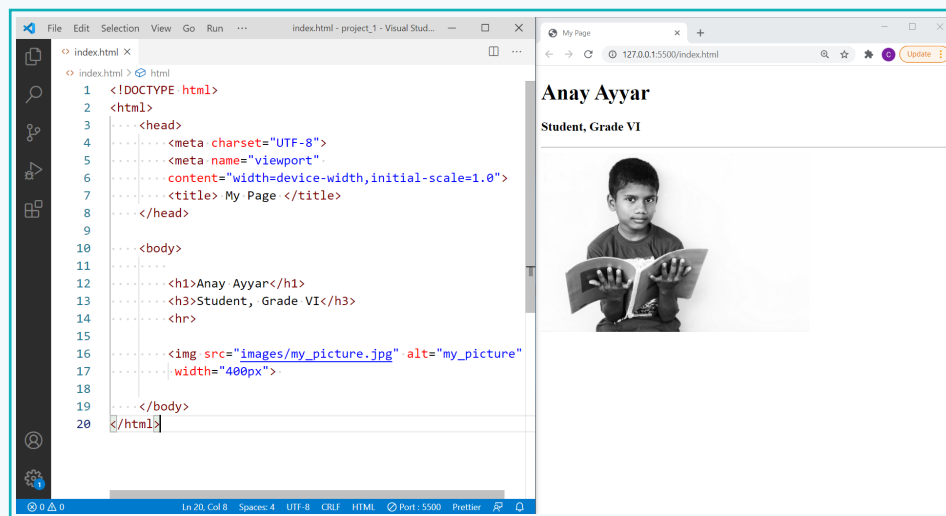
Earlier, we have created a simple personal website. In this lesson, we will add more functionalities to develop a personalised portfolio. We will add accomplishments, experiences, and contact details, among other information. Portfolios are a great way for job seekers to talk about their work experience and showcase their personal brand. From volunteering experiences to internships and personal work, a portfolio displays it all. When we create a portfolio, it's important to highlight our accomplishments, work experience, contact details etc.

Activity

Make a webpage showcasing your portfolio

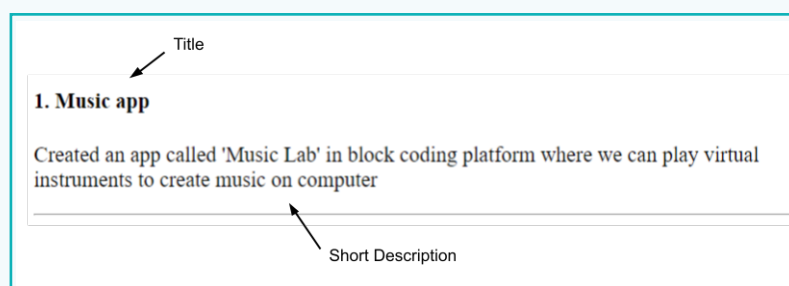
1 Create a basic webpage

- Open VS Code.
- Create a project folder and add it to the workspace in VS Code.
- Create a basic index.html webpage, with a heading and sub-heading about yourself. Add an image too.



Showcasing skills through projects or creative work done is the most important part of a portfolio. We will add a section with details about completed projects. Each work section will have 2 parts:

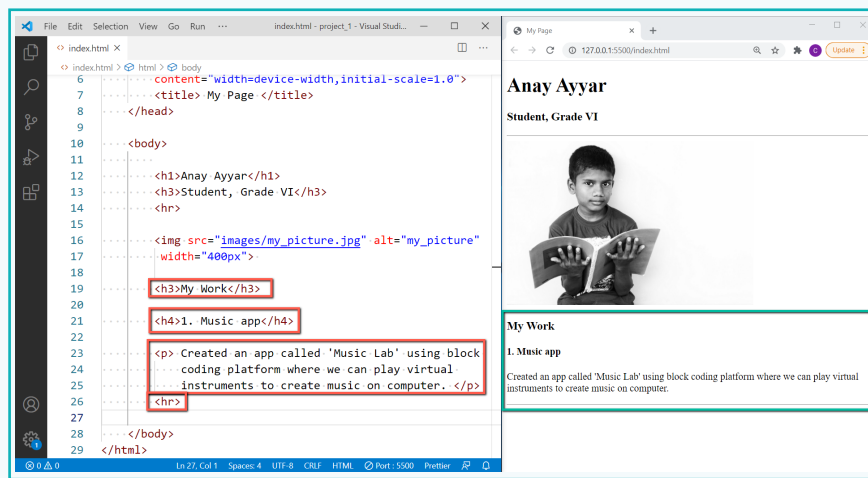
- Title of the work done
- A brief description of the work



Activity

2 Create 'My Work' section

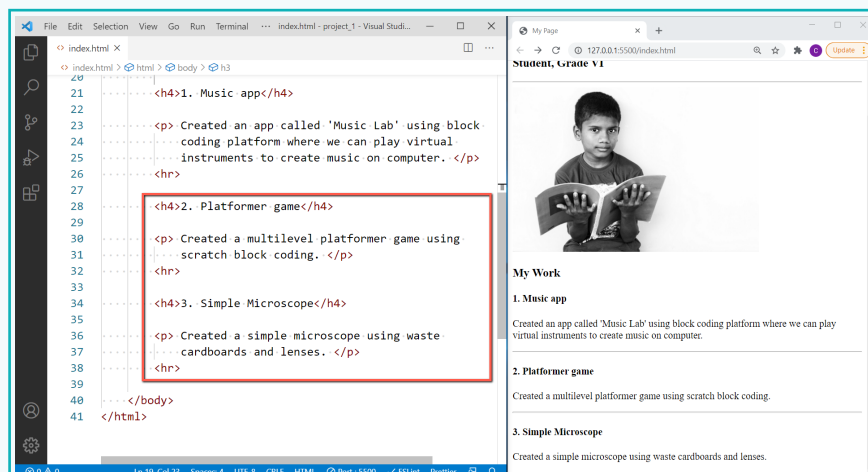
- Create a header for work using the `<h3>` tag. Name it 'My Work'.
- Add a sub-header using the `<h4>` tag as a title for project work.
- Add a brief description of the project using the `<p>` tag.



Recall:

The `<p>` tag is called the paragraph tag and is used to add a few sentences or paragraphs to a webpage.

Similarly, add more projects.

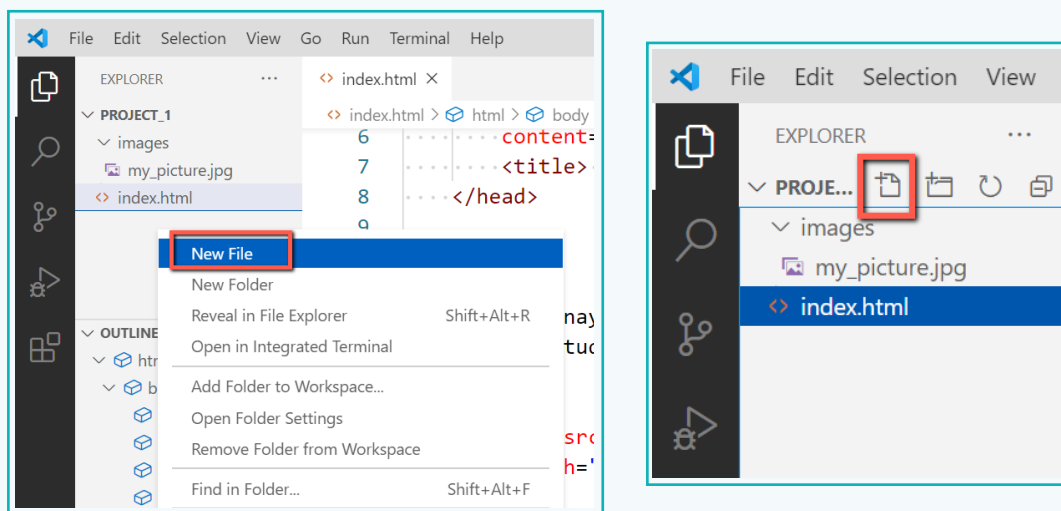


Activity

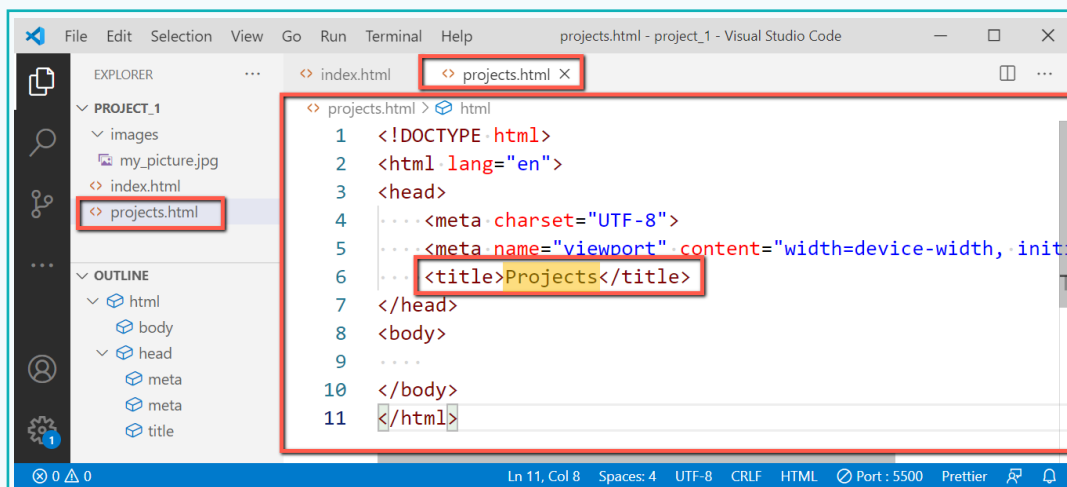
The most important feature of websites is the ability to have multiple pages where we can navigate from one page to another with the help of links. Let's create another webpage to showcase projects in detail.

3 Add a new webpage

Create a new file for a webpage by right-clicking in the explorer area. Select New File or click on the 'new file' symbol at the top of the explorer window.

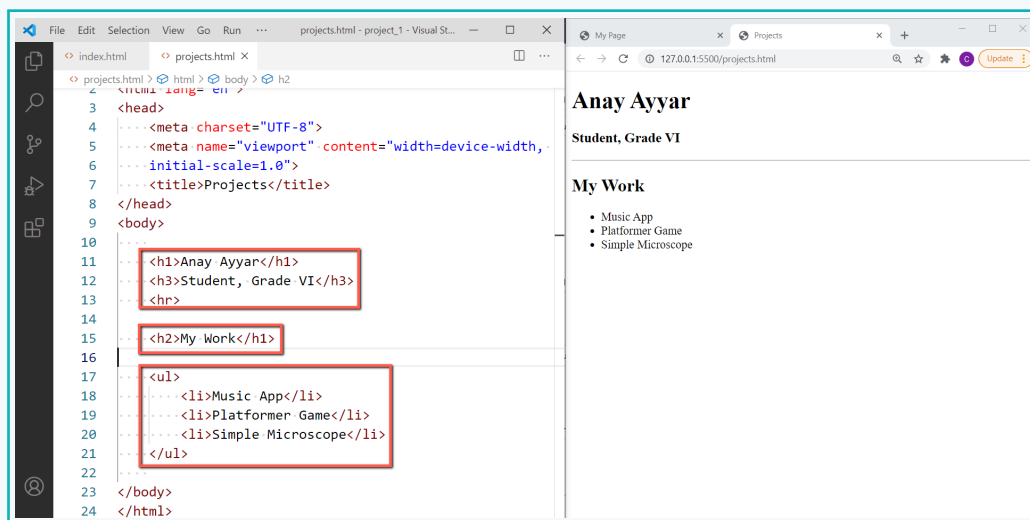


- Name the page project.html. A new webpage will be created in the same folder can be viewed in explorer.
- Type html and select html5 to insert a template code.
- Change the page name in the <title> tag to 'Projects'.



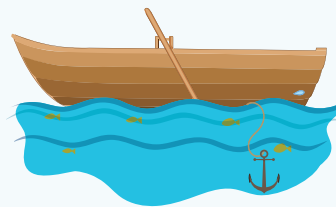
Activity

- Copy code for the main heading and sub-heading from index.html and paste it in the <body> tag of the new page.
- Add a heading with <h1> tag and text 'My Work' or 'Projects'.
- Add a list of project titles using an unordered list and list items tags.
- Press Alt + L + O to view the page output.



Now we can add a link to navigate from the main webpage to the projects page. Links can be created using anchor <a> tags.

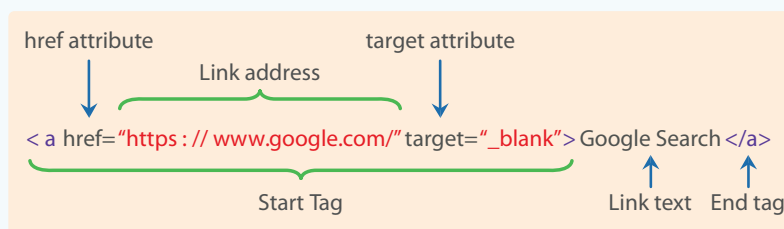
Anchor: An anchor is a metal object in a boat that acts as a hook and settles on the ocean bed so that boat does not drift away in the water.



A link on a webpage has two ends which act like anchors. One is at the link source and the other points to the destination. It can be any other web page or an element on a page. Links are created using anchor <a> tag.

Anchor element <a>: This is a tag used to create a hyperlink which links one page to another page or to an element on a page. 'href' attribute is used to add the address for the link. 'target' attribute specifies how to open the linked page.

For example:



Activity

Webpage output: A link is always displayed in a blue colour with an underline. If the link is clicked on, its colour changes to purple. This indicates that the page has been visited.

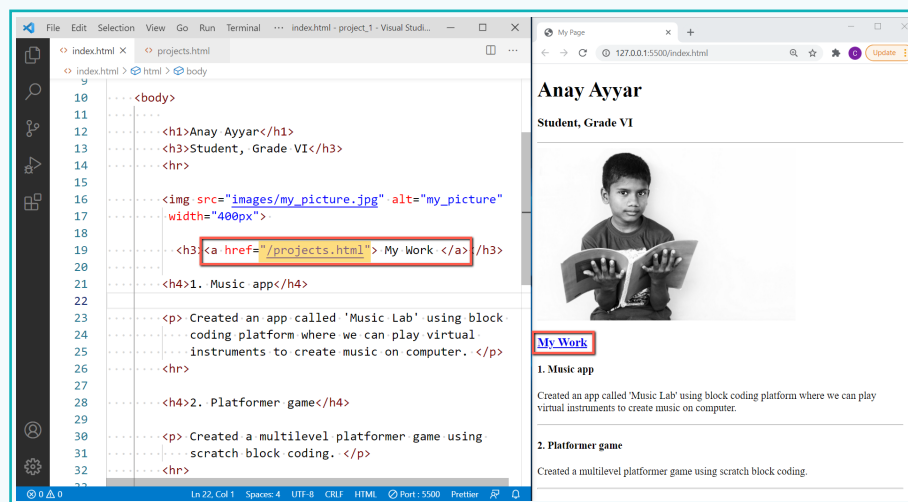
[Google Search](#)

[Google Search](#)

On the main page, we will add a link on the title for the projects section. This will link to the new webpage that has the project details.

4 Link to another page

- Insert section title text 'My Work' inside `<a>` `</a>` tags.
- Add the address of the project page `href="/ projects.html"` since the file is in the same folder as `index.html`.



Note:

It is important to start a URL with a backslash '/' so that the intended URL is returned instead of its parent or child.

If we click on the 'My Work' link, the projects page opens. To go back to the main page, for now, we can click on back. But we can also add a navigation bar to navigate through different pages and page sections.

5 Create the My Work Details section

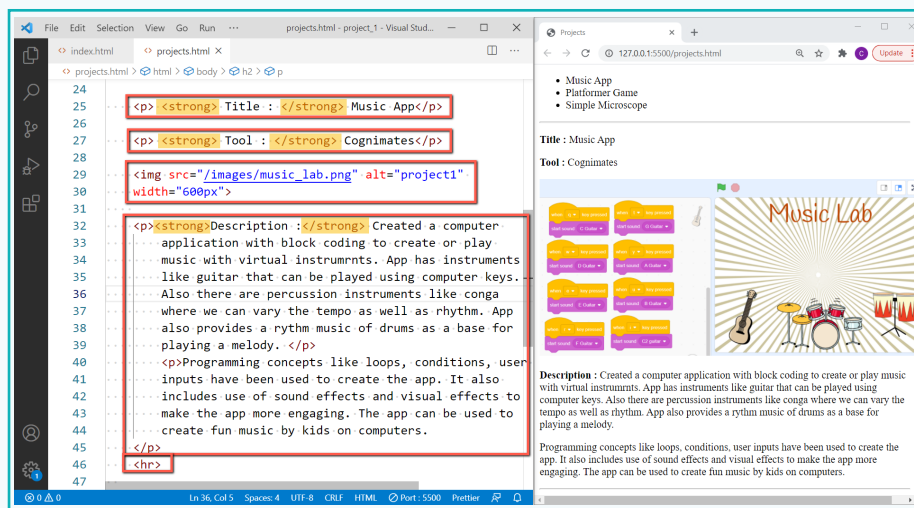
In this section, we will display details on the projects page about work completed. They will include the following:

- Title of the work done/project
- Tools used
- Images/video/audio of the project output
- Description of the project

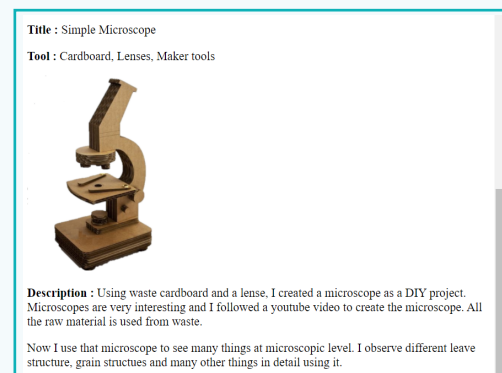
Activity

Creating the first work detail component

- Use a `<p>` tag to add a title, tools and a description for the projects.
- Encase the words with `<strong>` `</strong>` tags to display them as bold.
- Add an image of the project before the description to showcase the work.



Similarly, add descriptions and links for other projects. This can be done by duplicating the existing code and changing the contents of the tags. Also add link of a relevant website if necessary.



We have multiple projects to add, so we create a link that takes us to a specific part of the page instead of scrolling down on the page. For this, we use id attribute.

id attribute: Using an id attribute, we can assign a unique id to an HTML element. The id locates that specific element and uses it in links as the address.

For example: We want to add an id to a project title. We can add an id attribute to `<p>` starting tag as follows:

```
<p id="music_app"><strong>Title :</strong> Music App</p>
```

Activity

6 Add links to project titles

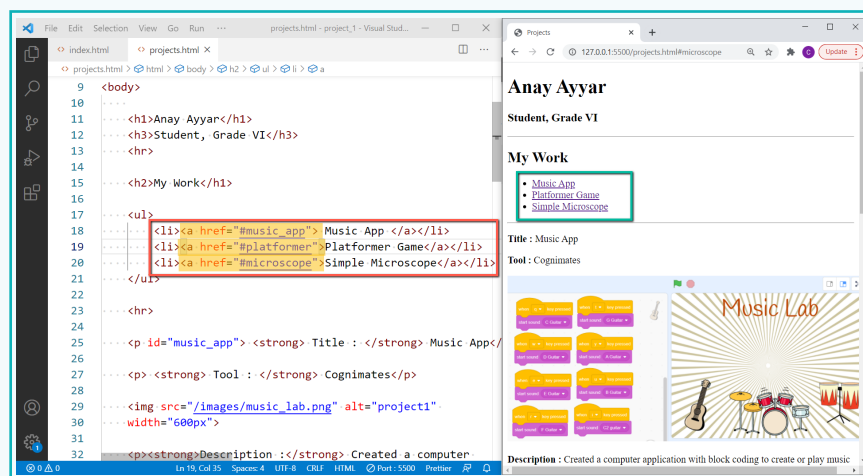
a. Add id attributes to <p> tags for the description titles of the other two projects.

```
<p id="platformer"><strong>Title :</strong> Platformer Game</p>
```

```
<p id="microscope"><strong>Title :</strong> Simple Microscope</p>
```

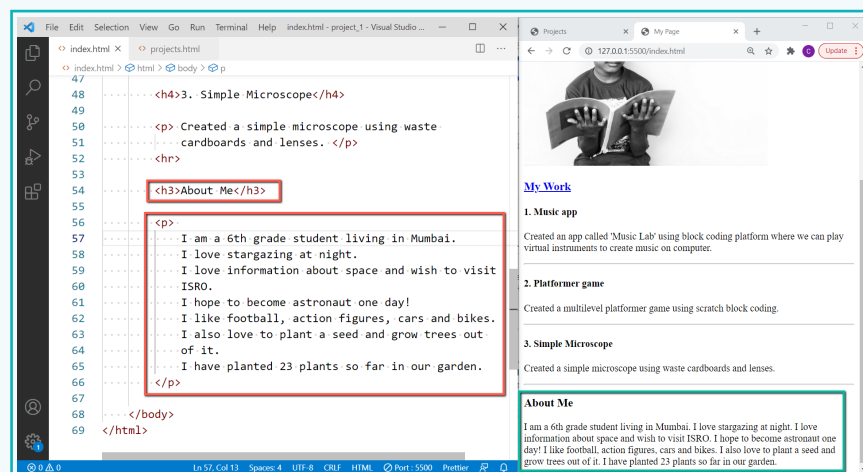
b. Insert text in tags for the project names in <a> tags.

c. Add the link address as "# + tag_name" in <a> tags.



Now we can navigate directly from the list of projects to project descriptions.

7 Add an "About Me" section after the projects section using <h3> and <p> tags



Activity

Note:

To differentiate between sections, we can add two horizontal lines in between them.

We can even directly link to any section of a new page. In this case, the link address consists of the page address followed by the id of the specific element. For example: the address for the platformer section on the projects page is href ="/projects.html#platformer"

Exercise

 **Tick mark ☒ the correct answer**

1 _____ element is used to link text or images to a new webpage or to a specific section of the page.

a. Form ☐

b. Anchor ☐

c. Input ☐

d. Button ☐

2 _____ attribute contains the address where an anchor <a> tag links to.

a. Target ☐

b. Class ☐

c. Id ☐

d. href ☐

 **State whether True or False**

3 We can link to a specific element on the webpage using an id attribute.

 _____

4 A visited link is coloured blue with an underline whereas a fresh link is shown as purple with an underline.

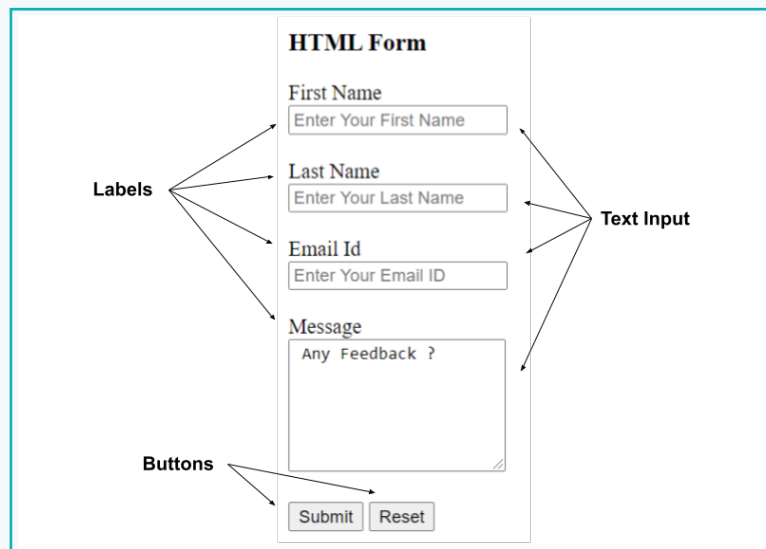
 _____

Activity

Each portfolio has a section with contact details and sometimes a contact form. The form can be used to send a message or email to the portfolio owner directly from the website.

HTML Forms:

HTML forms are an essential part of any website that wants to interact with its viewers (people visiting a website). A form is used to collect information from its users.



The diagram illustrates the components of an HTML form. It shows a form titled "HTML Form" with the following elements:

- Labels:** Four labels are listed on the left: "First Name", "Last Name", "Email Id", and "Message". Arrows point from each label to its corresponding input field.
- Text Input:** Four text input fields are shown on the right, each containing placeholder text: "Enter Your First Name", "Enter Your Last Name", "Enter Your Email ID", and "Any Feedback ?". Arrows point from each input field to the "Text Input" label.
- Buttons:** Two buttons, "Submit" and "Reset", are located at the bottom of the form. An arrow points from the "Buttons" label to these buttons.

Creating Forms using HTML:

Form element: To create a form in HTML, we will use the `<form> .... </form>` tag. It acts like a container inside which we can add form elements like text-boxes, labels, buttons, checkboxes etc. for input.

The `<input>` element is the most used element and is used to create different types of input options like text-boxes, radio buttons, checkboxes, drop-down lists and such.

The **`<label>`** element is used to create labels for input elements.

Bytes

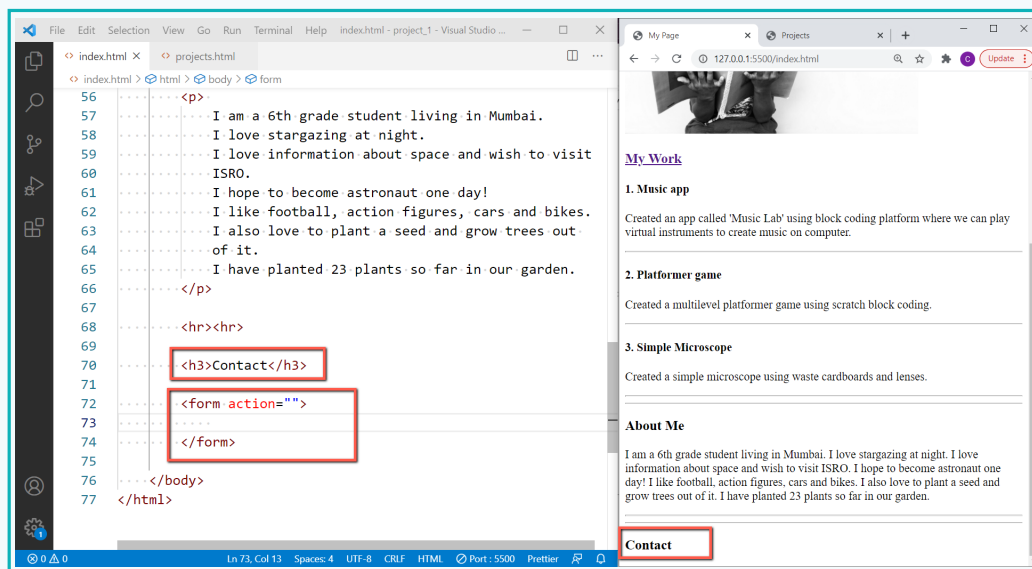
A dummy placeholder text is commonly used in webpages to demonstrate a document structure and layout. The complete text can be copied from here:

"Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip..."

Activity

8 Now that we know more about forms, we will create a Contact Form for our web site.

- Create a heading using the `<h3>` tag
- Type “form” and press enter to add the `<form>` tag



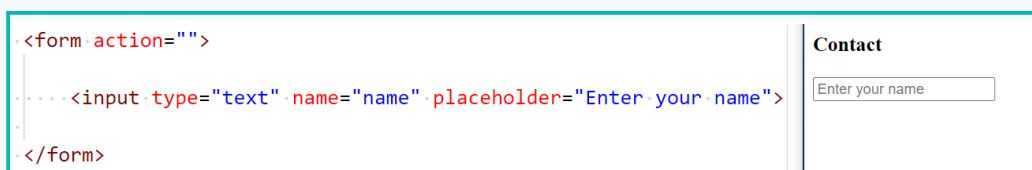
Notice that no output is visible for our `<form>` tag. This is because we have not added any form elements yet.

- Create a text box for the viewer to enter their name with the following code:

```
<input type="text" name="name" placeholder="Enter your name">
```

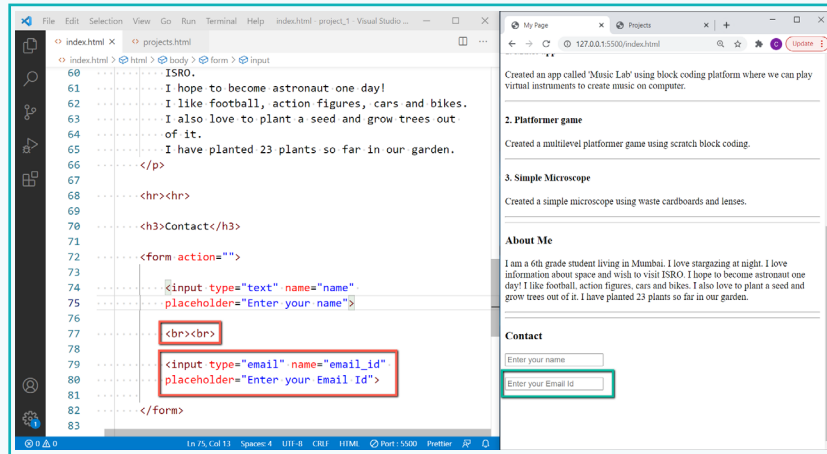
This element has 3 attributes:

- The **type** attribute decides the type of input.
- The **name** attribute is used to give the textbox a name.
- The **placeholder** attribute is used to display text inside the text box.



Activity

- d. Add a `<br>` tag to move content to the next line.
- e. Similarly, create another text box to enter an email id.



Note:

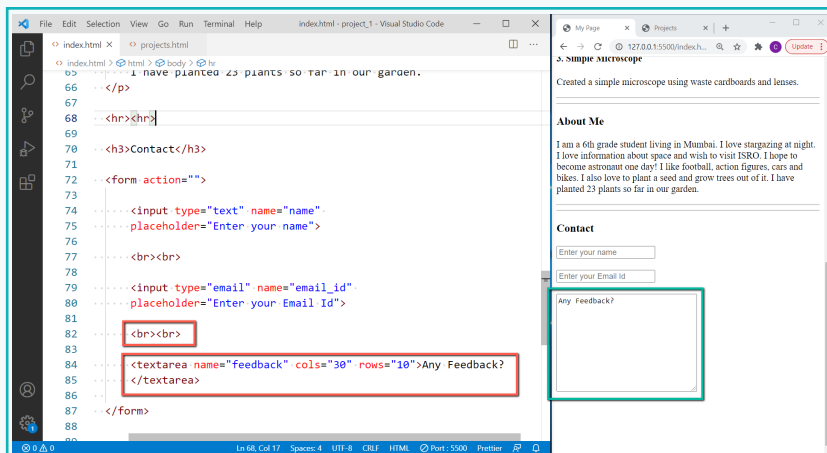
Some part of the tag can be moved to the next line.

`<br>` tag, known as the Break tag, is used to insert line breaks between html elements.

We will now create a text area for visitors to leave their feedback. A **text-area** is similar to a textbox, where users can enter multiple lines of text.

- f. Add a `<br>` tag to add a text area after leaving two empty lines.
- g. Add a `<textarea>` tag to create a large text area.
- h. Write “Any Feedback?” in between `<textarea>` and `</textarea>`.

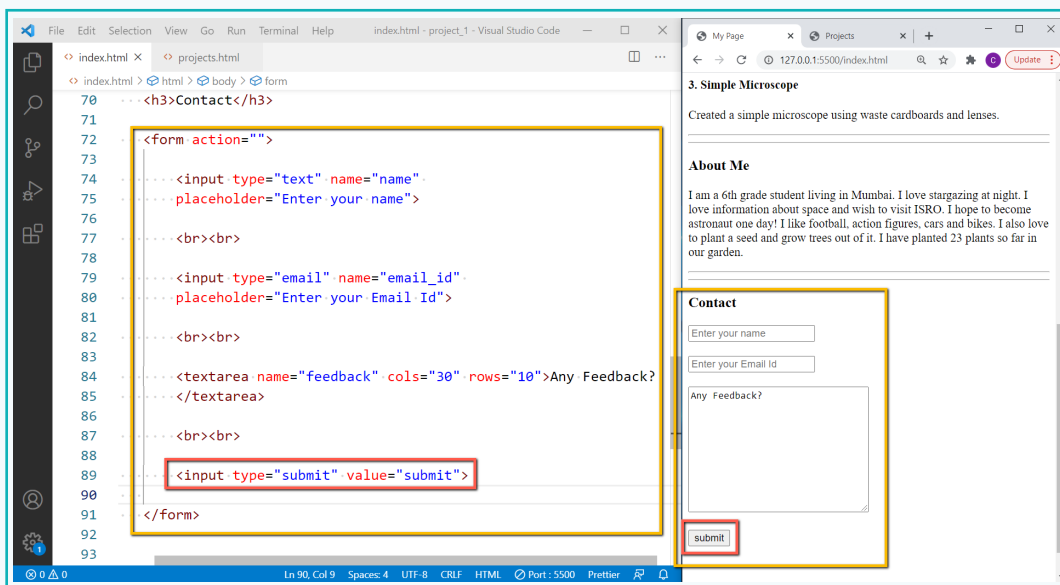
The “rows” attribute specifies the height of the text area and “column” specifies the width.



Activity

To submit the form, we can create a Submit button. We can create a button using `<input>` tag with **type =“submit”**.

- i. Add a button using an `<input>` tag.
- j. The 'value' attribute specifies the value of an `<input>` element. “submit” type defines the text on the button.



The contact form is complete with the code.

Exercise

 **Tick mark ☒ the correct answer**

5 _____ element is used to add a small text input window for name or email

a. Form ☐

b. Input ☐

c. Textbox ☐

d. Button ☐

6 <input> tag is used with type _____ to add button.

a. Textarea ☐

b. Checkbox ☐

c. Text ☐

d. Submit ☐

7 _____ attribute is used to show light coloured text in a textbox as an indicator of its purpose.

a. Id ☐

b. Placeholder ☐

c. Type ☐

d. Name ☐

 **State whether True or False**

8 <textarea> </textarea> tag is used to add a large text input area.

 _____

9 Row and column attributes in a 'textarea' element decide the height and width of the text input box respectively.

 _____

Activity

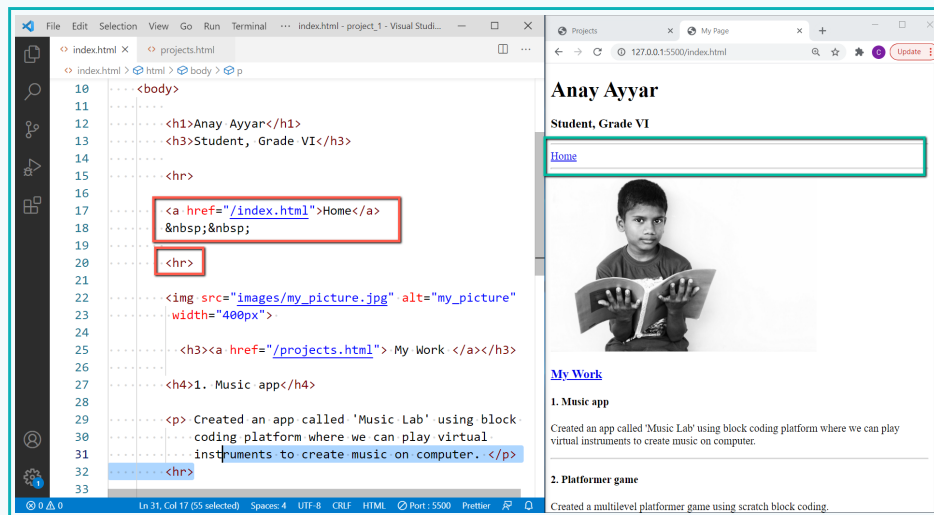
Navbar: Navigation bars are an important part of a website. Navigation bar or Navbar contains names of important sections and links us to those sections.

For example: Usually, navigation bars are at the top of a webpage. All pages of a website contain a navbar to navigate through the website. Here is a simple HTML navbar.

[Home](#) [Projects](#) [About me](#) [Contact](#)

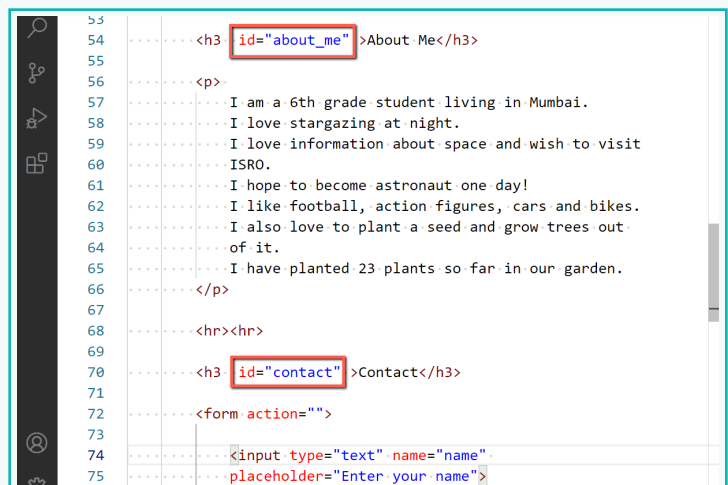
9 Add a navigation bar

- Add an anchor tag `<a href="">` to add a link name 'Home'. Link the name to the address of the main page with `href="/index.html"`.
- Add `' '` character set twice to add space after the link.
- Add a horizontal line.



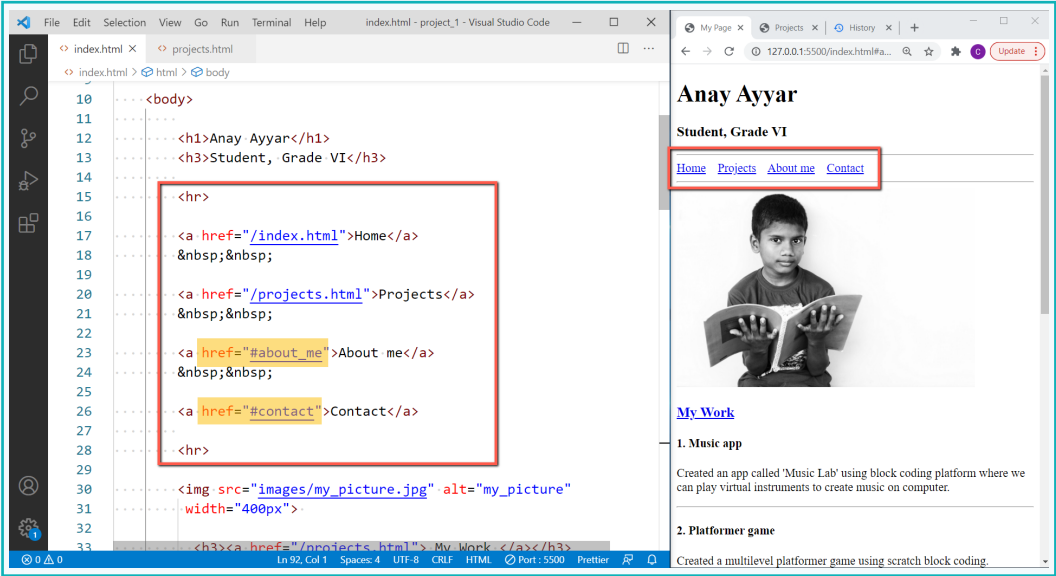
Similarly, add more navbar links like Projects, About me and Contact.

- Add an id to `<h3>` tags of 'About me' and 'Contact' section.

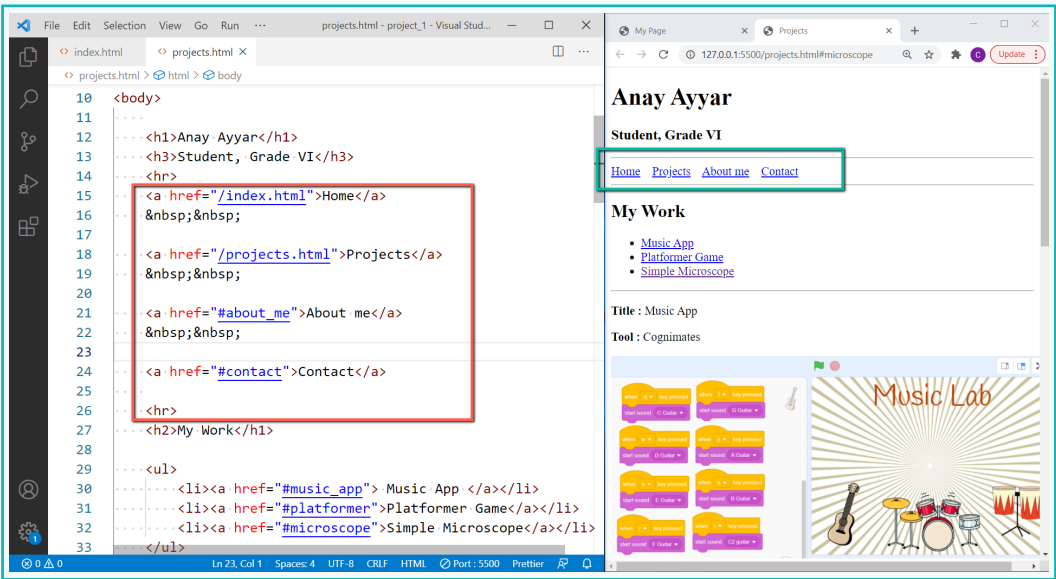


Activity

- e. Add the page address href="/projects.html" to the projects link.
- f. Add addresses to the links 'about me' and 'contact' in navbar using id.



- g.** Copy the same navbar code and paste it on the projects page at the same position.



Our portfolio website is complete!

Exercise

State whether True or False

10 Navbar is used to navigate through different webpages and important sections of a website.

 _____

11 Navbar can be located anywhere but is usually at the top of the website.

 _____

In this lesson, we explored various elements of a portfolio and created our own portfolio website. A portfolio is a set of information and examples of our work and achievements that we use when entering competitions or applying for work. Using anchor `<a>` tags, we added links to navigate from the main webpage to the project page. We also added a navigation bar to navigate through different webpages and sections of a page.

In a portfolio, always show the best of what you have, you don't need to show it all. People have short attention spans, so be thoughtful with what you include.



Tech Flash

- **Portfolio** : A portfolio is a collection of works or documents to demonstrate a person's skills and accomplishments.
- **Link** : A link can be a text or an image. When clicked, it takes us to a linked web page or a specific section of a webpage. An anchor element is used to add links in HTML.
- **id** : id is an attribute that specifies a unique id for an HTML element. It can be used as an address for that element in a link.
- **Form** : Form is a section that provides input options like text input window, checkbox, buttons etc. It is used to interact with website visitors.
- **Navbar** : Navbar directly links us to webpages or sections of pages on a website.